ClipSAT : Multivariable Calculus Partials, Multiple Integrals & Vector Calcu

Unit 6.01: Vector Fields, Conservative Fields & Line Integrals Name: _____

Q1. Consider the vector field mathbf{F}x,y = langle 3x, 2y rangle . Evaluate the vector field at the point 2, 4 .

[ ] A: langle 6, 8 rangle
[ ] B: langle 8, 6 rangle
[ ] C: langle 5, 6 rangle
[ ] D: langle 6, 5 rangle

Q2. Consider the vector field mathbf{F}x, y = langle 2x, 3y rangle . Which of the following options best represents the vector field at the point 1, 1?

[ ] A: langle 2, 3 rangle
[ ] B: langle 1, 1 rangle
[ ] C: langle 4, 6 rangle
[ ] D: langle 2, 6 rangle

Q3. Consider the vector field mathbf{F}x,y = langle 2x+y, x-3y rangle . What is the value of the vector field at the point 3, -1 ?

[ ] A: langle 5, 6 rangle
[ ] B: langle 5, 0 rangle
[ ] C: langle 7, 6 rangle
[ ] D: langle 7, 0 rangle

Q4. Evaluate int_C x^{2},ds where C is the segment from left 0, 0right to left 5, 12right .

[ ] A: frac{25}{3}
[ ] B: frac{4225}{3}
[ ] C: frac{328}{3}
[ ] D: frac{325}{3}

Q5. Which vector field is conservative on all of mathbb R^2 ?

[ ] A: langle e^{y}, x e^{y} rangle
[ ] B: langle x y, x rangle
[ ] C: langle x + y, - x + y rangle
[ ] D: langle x^{2} y, x y^{2} rangle

Q6. Evaluate int_Cmathbf Fcdot dmathbf r for mathbf F=langle y^{2}, 2 x rangle along the segment from left -2, -2right to left -2, 3right .

[ ] A: 20
[ ] B: -19
[ ] C: -40
[ ] D: -20

Q7. Evaluate int_C x + y,ds where C is the segment from left 0, 0right to left 4, 3right .

[ ] A: frac{35}{2}
[ ] B: frac{7}{2}
[ ] C: frac{175}{2}
[ ] D: frac{37}{2}

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Q8. Evaluate int_Cmathbf Fcdot dmathbf r for mathbf F=langle x y, y rangle along the segment from left 1, 1right to left 1, -2right .

[ ] A: - frac{3}{2}

[ ] B: frac{5}{2}

[ ] C: frac{3}{2}

[ ] D: 3